Noah Lowery

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Education

California State University San Marcos, San Marcos, CA
Bachelor of Science in Mathematics Graduated May 2025
Relevant Coursework: Financial Mathematics, Machine Learning in Finance, Optimization, Stochastic Processes, Data Structures and Algorithms

Professional Experience

Peer Leader, Learning and Tutoring Services *CSU San Marcos, Apr 2023 – May 2025*

- Facilitated weekly study sessions for 30+ students in calculus and linear algebra.
- Increased facility utilization through engagement-focused workshops.
- Acted as liaison between faculty and students, guiding peer tutors and sharing feedback.

Radical Cubic Extensions Research *CSU San Marcos, May 2024 – Aug 2024*

- Conducted research on classifying radical cubic field extensions over polynomial rings using algebraic methods.
- Presented results to university leadership, math faculty, and conference audiences with varying levels of technical background.

Research and Awards

Brownian Motion & First-Exit Time Modeling *Senior Seminar Project*

- Built a Python simulation to model first-exit times and profit distributions using Brownian motion and utility theory.
- Applied NumPy and pandas to generate and analyze stochastic time series data; visualized model outcomes for both technical and general audiences.
- Explored real-world implications for timing strategy in investment decisions and risk-based exit models.

Wolfram Award – Technical Integration in Mathematics *CSU San Marcos*

- Recognized for incorporating Python and SageMath into math coursework and research, emphasizing clarity and technical application.
- Demonstrated ability to connect abstract mathematical theory with hands-on modeling through research and presentations.

Skills

- **Technical:** Python, NumPy, Excel, SageMath, LaTeX, R
- **Quantitative:** Simulation modeling, stochastic processes, time-to-event analysis, first-exit models, risk analysis
- **Communication:** Technical writing, academic presentations, peer tutoring; experience translating technical findings for general audiences